

LAB 9 - Candy Mini-Project

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```
candy_file <- "candy-data.csv"

candy = read.csv(candy_file, row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

2.1 What is in the dataset?

Q1- How many different candy types are in this dataset?

There is 85 different types of candy in this dataset.

```
nrow(candy)
```

```
[1] 85
```

Q2- How many fruity canfy types are there in the dataset?

There is 38 fruity candy types in the dataset.

```
sum(candy$fruity)
```

```
[1] 38
```

2.2 What is your favorite candy?

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
      winpercent
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is its winpercent value?

```
library(dplyr)

candy |>
  filter(row.names(candy)=="Snickers") |>
  select(winpercent)
```

```
      winpercent
Snickers  76.67378
```

Q4. What is the winpercent value for “Kit Kat”?

```
library(dplyr)

candy |>
  filter(row.names(candy)=="Kit Kat") |>
  select(winpercent)
```

```
      winpercent
Kit Kat  76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
library(dplyr)

candy |>
  filter(row.names(candy)=="Tootsie Roll Snack Bars") |>
  select(winpercent)
```

```
      winpercent
Tootsie Roll Snack Bars  49.6535
```

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes winpercent is on a different scale that is roughly 0–100 compared to most other columns, which are binary 0/1 variables.

Q7. What do you think a zero and one represent for the candy\$chocolate column?

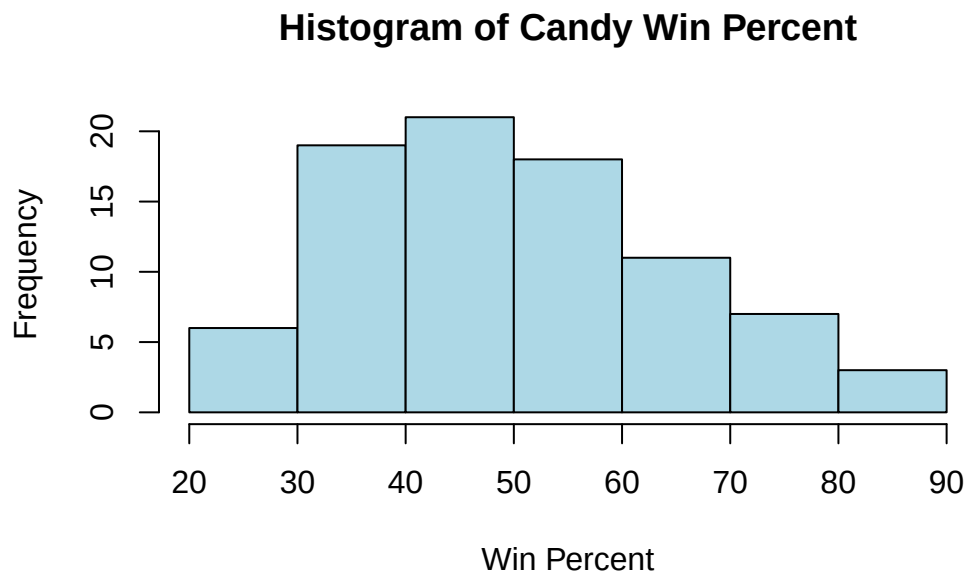
In the chocolate column, 1 means the candy contains chocolate and 0 means it does not.

```
table(candy$chocolate)
```

```
0 1
48 37
```

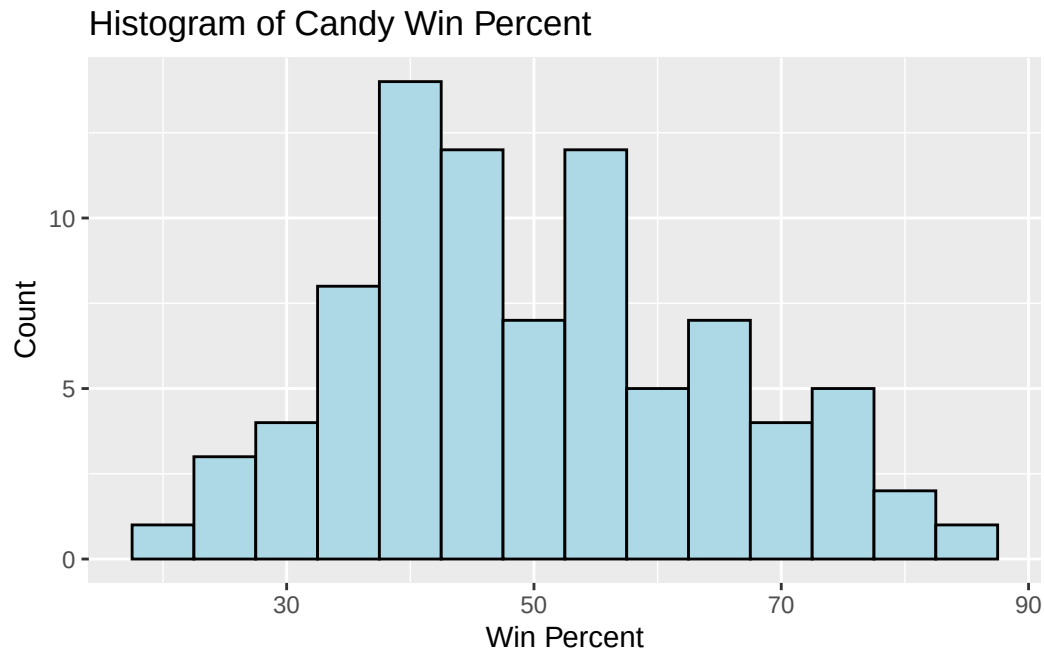
Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
# Base R histogram of winpercent
hist(candy$winpercent,
     main = "Histogram of Candy Win Percent",
     xlab = "Win Percent",
     col = "lightblue",)
```



```
library(ggplot2)

# ggplot2 histogram of winpercent
ggplot(candy, aes(x = winpercent)) +
  geom_histogram(binwidth = 5, fill = "lightblue", color = "black") +
  labs(title = "Histogram of Candy Win Percent",
       x = "Win Percent",
       y = "Count")
```



Q9. Is the distribution of winpercent values symmetrical?

No, the distribution is slightly skewed not perfectly symmetrical.

Q10. Is the center of the distribution above or below 50%?

```
# Calculate mean winpercent  
mean(candy$winpercent)
```

```
[1] 50.31676
```

The center is below 50 %

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
# Mean winpercent for chocolate vs fruity candies  
mean(candy$winpercent[as.logical(candy$chocolate)])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[as.logical(candy$fruity)])
```

```
[1] 44.11974
```

Chocolate candy is ranked higher on average than fruity candy.

Q12. Is this difference statistically significant?

```
# Perform t-test comparing chocolate vs fruity candies
t.test(candy$winpercent[as.logical(candy$chocolate)],
       candy$winpercent[as.logical(candy$fruity)])
```

Welch Two Sample t-test

```
data: candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical(candy$fruity)]
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Yes, the difference is statistically significant with a very small p-value of 2.87e-08.

```
library(dplyr)

# Bottom 5 (least liked candies)
candy |>
  arrange(winpercent) |> # sort from lowest to highest
  head(5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

crispedrice wafer hard bar pluribus sugarpercent pricepercent

Nik L Nip	0	0	0	1	0.197	0.976
Boston Baked Beans	0	0	0	1	0.313	0.511
Chiclets	0	0	0	1	0.046	0.325
Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511

winpercent

Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

```
# Top 5 (most liked candies)
candy |>
  arrange(desc(winpercent)) |> # sort from highest to lowest
  head(5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

	crisped	rice	wafer	hard bar	pluribus	sugar	percent
Reese's Peanut Butter cup		0	0	0	0		0.720
Reese's Miniatures		0	0	0	0		0.034
Twix		1	0	1	0		0.546
Kit Kat		1	0	1	0		0.313
Snickers		0	0	1	0		0.546

	price	percent	winpercent
Reese's Peanut Butter cup	0.651		84.18029
Reese's Miniatures	0.279		81.86626
Twix	0.906		81.64291
Kit Kat	0.511		76.76860
Snickers	0.651		76.67378

```
# Bottom 5
head(candy[order(candy$winpercent), ], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0

Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0

	crispedricewafer	hard	bar	pluribus	sugarpercent	pricepercent
Nik L Nip	0	0	0	1	0.197	0.976
Boston Baked Beans	0	0	0	1	0.313	0.511
Chiclets	0	0	0	1	0.046	0.325
Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

```
# Top 5
head(candy[order(-candy$winpercent), ], 5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Reese's Peanut Butter cup	1	0	0	1	0
Reese's Miniatures	1	0	0	1	0
Twix	1	0	1	0	0
Kit Kat	1	0	0	0	0
Snickers	1	0	1	1	1

	crispedricewafer	hard	bar	pluribus	sugarpercent
Reese's Peanut Butter cup	0	0	0	0	0.720
Reese's Miniatures	0	0	0	0	0.034
Twix	1	0	1	0	0.546
Kit Kat	1	0	1	0	0.313
Snickers	0	0	1	0	0.546

	pricepercent	winpercent
Reese's Peanut Butter cup	0.651	84.18029
Reese's Miniatures	0.279	81.86626
Twix	0.906	81.64291
Kit Kat	0.511	76.76860
Snickers	0.651	76.67378

Q13. What are the five least liked candy types in this set?

The five least liked candy types in this set are Nik L Nip, Boston Baked Beans Chiclets, Super Bubble, and Jawbusters.

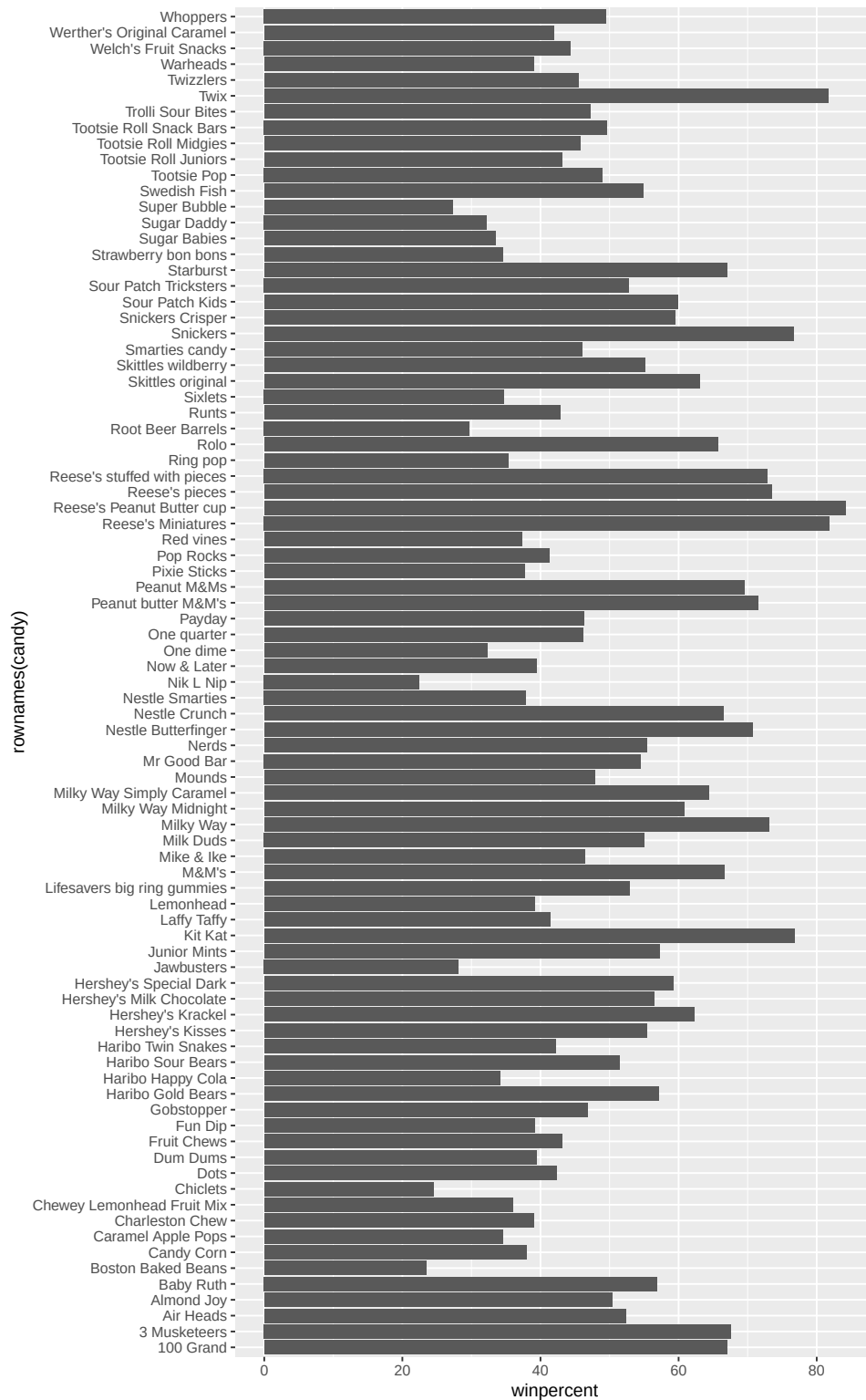
Q14. What are the top 5 all time favorite candy types out of this set?

The five most liked candy types in this set are Reese's Peanut Butter Cup Reese's Miniatures, Twix, Kit Kat, and Snickers

Q15. Make a first barplot of candy ranking based on winpercent values.

```
library(ggplot2)

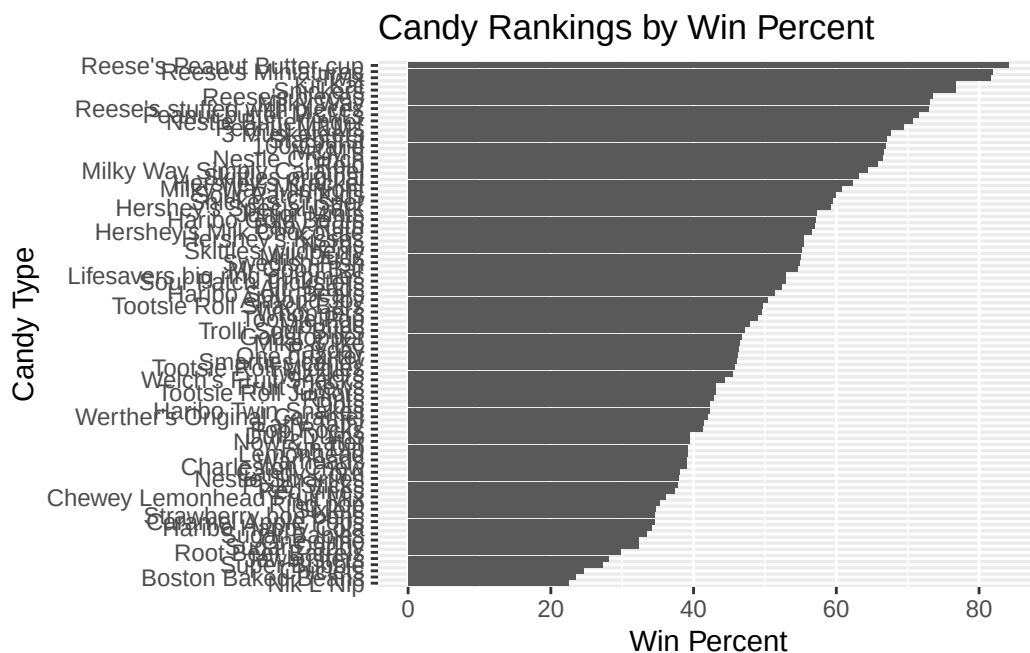
# Basic barplot of winpercent vs candy names
ggplot(candy) +
  aes(x = winpercent, y = rownames(candy)) + # map winpercent to x-axis, candy names to y-axis
  geom_col() # use geom_col() to create bars with actual values
```



Q16 This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

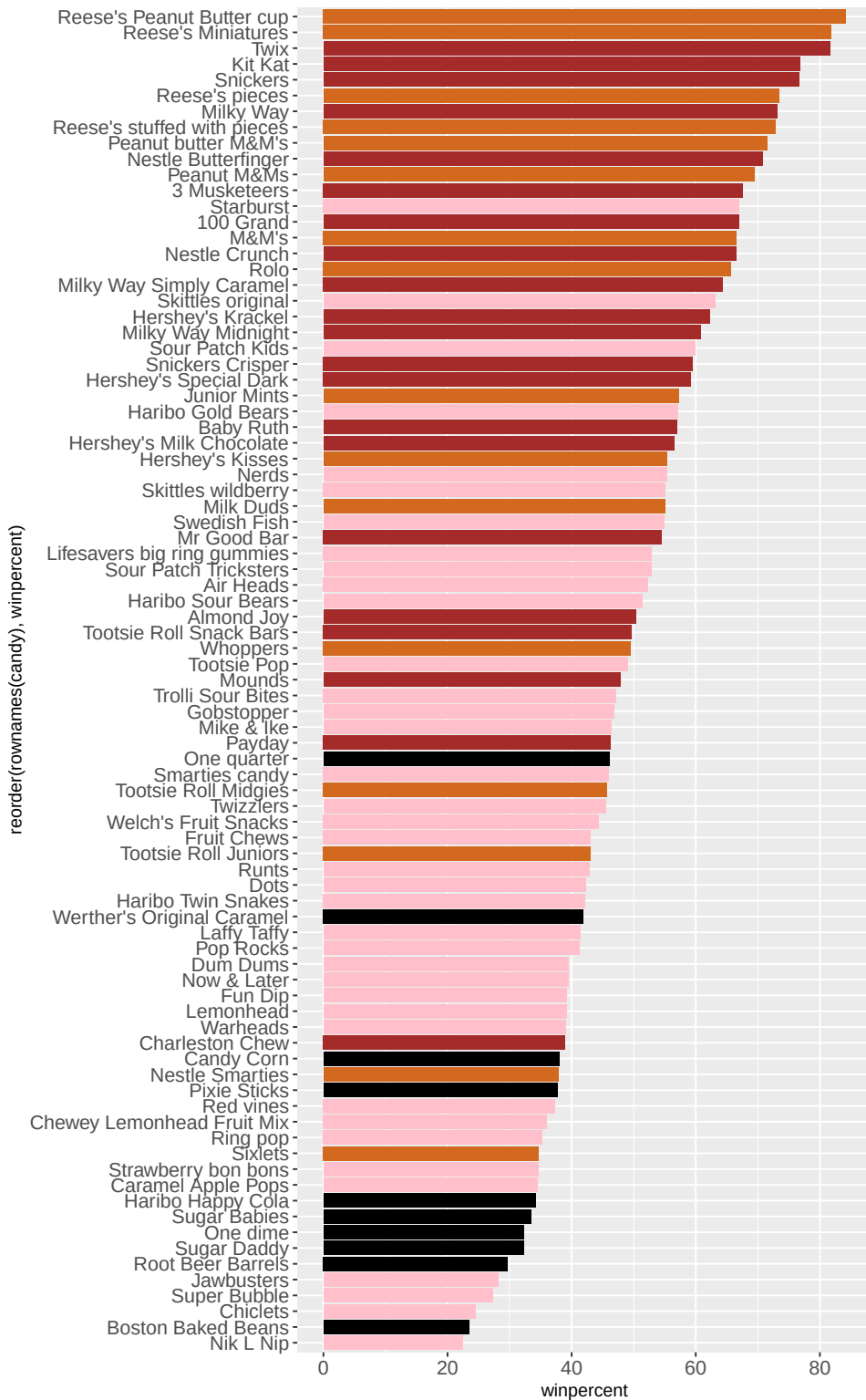
```
library(ggplot2)

# Barplot with candies sorted by winpercent
ggplot(candy) +
  aes(x = winpercent,
      y = reorder(rownames(candy), winpercent)) + # reorder candy names by winpercent
  geom_col() +
  labs(title = "Candy Rankings by Win Percent",
       x = "Win Percent",
       y = "Candy Type")
```



```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"
```

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols) +
  theme(axis.text = element_text(size = 12))
```



Q17. What is the worst ranked chocolate candy?

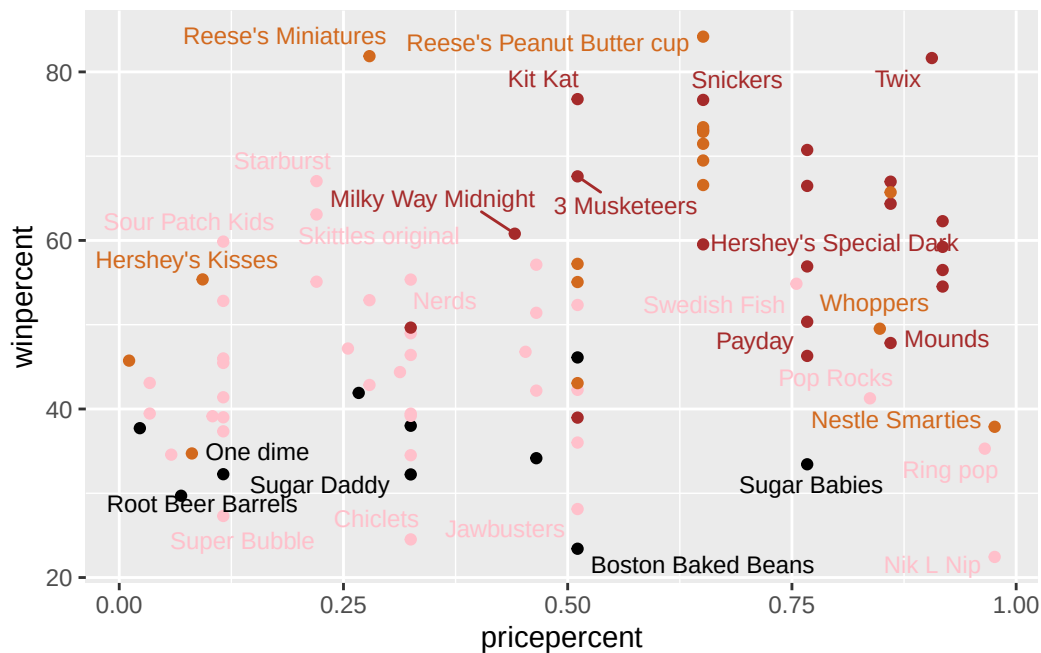
The worst ranked chocolate candy bar was Nik L Nip

Q18. What is the best ranked fruity candy?

The nest ranked fruity bar was Reeses penut butter cup.

```
library(ggrepel)

# Plot of winpercent vs pricepercent
ggplot(candy) +
  aes(x = pricepercent, y = winpercent, label = rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size = 3.3, max.overlaps = 5)
```



****Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?***

The candy with the highest winpercent relative to its price is Reese's Miniatures

```
# Create value score: popularity per price
candy$value_score <- candy$winpercent / candy$pricepercent

# Find the best "bang for your buck"
candy[order(-candy$value_score), c("winpercent", "pricepercent")]
```

	winpercent	pricepercent
Tootsie Roll Midgies	45.73675	0.011
Pixie Sticks	37.72234	0.023
Fruit Chews	43.08892	0.034
Dum Dums	39.46056	0.034
Strawberry bon bons	34.57899	0.058
Hershey's Kisses	55.37545	0.093
Sour Patch Kids	59.86400	0.116
Sour Patch Tricksters	52.82595	0.116
Root Beer Barrels	29.70369	0.069
Sixlets	34.72200	0.081
Smarties candy	45.99583	0.116
Twizzlers	45.46628	0.116
Lemonhead	39.14106	0.104
Laffy Taffy	41.38956	0.116
Warheads	39.01190	0.116
Red vines	37.34852	0.116
Starburst	67.03763	0.220
Reese's Miniatures	81.86626	0.279
Skittles original	63.08514	0.220
One dime	32.26109	0.116
Skittles wildberry	55.10370	0.220
Super Bubble	27.30386	0.116
Lifesavers big ring gummies	52.91139	0.279
Trolli Sour Bites	47.17323	0.255
Nerds	55.35405	0.325
Werther's Original Caramel	41.90431	0.267
Runts	42.84914	0.279
Tootsie Roll Snack Bars	49.65350	0.325
Tootsie Pop	48.98265	0.325
Kit Kat	76.76860	0.511
Mike & Ike	46.41172	0.325
Welch's Fruit Snacks	44.37552	0.313
Milky Way Midnight	60.80070	0.441
3 Musketeers	67.60294	0.511
Reese's Peanut Butter cup	84.18029	0.651

Haribo Gold Bears	57.11974	0.465
Now & Later	39.44680	0.325
Fun Dip	39.18550	0.325
Snickers	76.67378	0.651
Candy Corn	38.01096	0.325
Reese's pieces	73.43499	0.651
Milky Way	73.09956	0.651
Junior Mints	57.21925	0.511
Reese's stuffed with pieces	72.88790	0.651
Haribo Sour Bears	51.41243	0.465
Peanut butter M&M's	71.46505	0.651
Milk Duds	55.06407	0.511
Peanut M&Ms	69.48379	0.651
Caramel Apple Pops	34.51768	0.325
Gobstopper	46.78335	0.453
Air Heads	52.34146	0.511
M&M's	66.57458	0.651
Sugar Daddy	32.23100	0.325
Nestle Butterfinger	70.73564	0.767
Snickers Crisper	59.52925	0.651
Haribo Twin Snakes	42.17877	0.465
One quarter	46.11650	0.511
Twix	81.64291	0.906
Nestle Crunch	66.47068	0.767
Tootsie Roll Juniors	43.06890	0.511
Dots	42.27208	0.511
100 Grand	66.97173	0.860
Rolo	65.71629	0.860
Charleston Chew	38.97504	0.511
Chiclets	24.52499	0.325
Milky Way Simply Caramel	64.35334	0.860
Baby Ruth	56.91455	0.767
Haribo Happy Cola	34.15896	0.465
Swedish Fish	54.86111	0.755
Chewey Lemonhead Fruit Mix	36.01763	0.511
Hershey's Krackel	62.28448	0.918
Almond Joy	50.34755	0.767
Hershey's Special Dark	59.23612	0.918
Hershey's Milk Chocolate	56.49050	0.918
Payday	46.29660	0.767
Mr Good Bar	54.52645	0.918
Whoppers	49.52411	0.848
Mounds	47.82975	0.860

Jawbusters	28.12744	0.511
Pop Rocks	41.26551	0.837
Boston Baked Beans	23.41782	0.511
Sugar Babies	33.43755	0.767
Nestle Smarties	37.88719	0.976
Ring pop	35.29076	0.965
Nik L Nip	22.44534	0.976

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

the top 5 most expensive candies are Nik L Nip, Nestle Smarties, Ring Pop Hershey's Krackel and Hershey's Milk Chocolate.

```
# Order by most expensive first
ord <- order(candy$pricepercent, decreasing = TRUE)

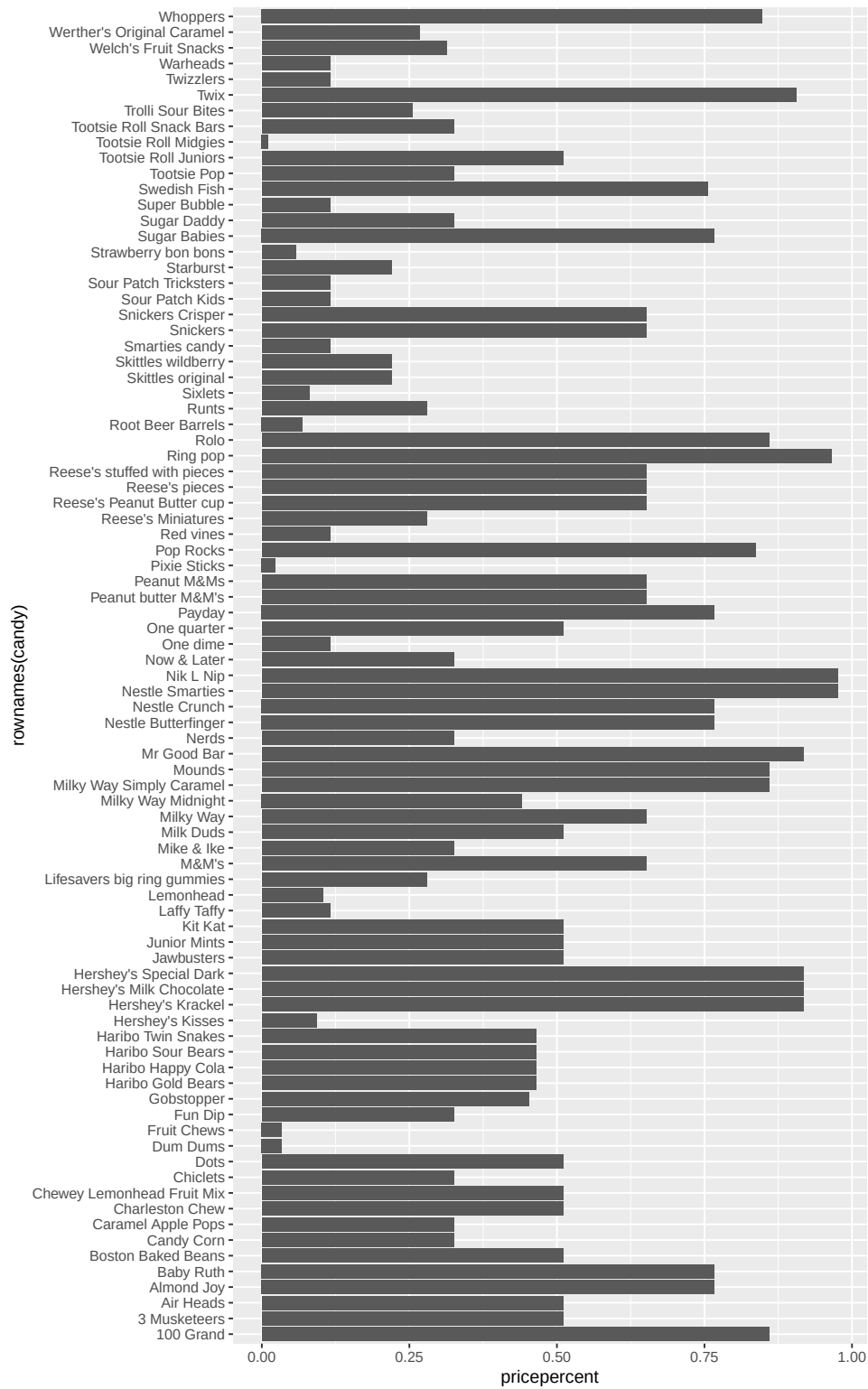
# Top 5 most expensive candies
top5_expensive <- candy[ord, c("pricepercent", "winpercent")][1:5, ]
top5_expensive
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

Q21. Make a barplot again with `geom_col()` this time using `pricepercent` and then improve this step by step, first ordering the x-axis by value and finally making a so called “dot chat” or “lollipop” chart by swapping `geom_col()` for `geom_point()` + `geom_segment()`.

```
library(ggplot2)

# Basic barplot of pricepercent
ggplot(candy) +
  aes(x = pricepercent, y = rownames(candy)) +
  geom_col()
```



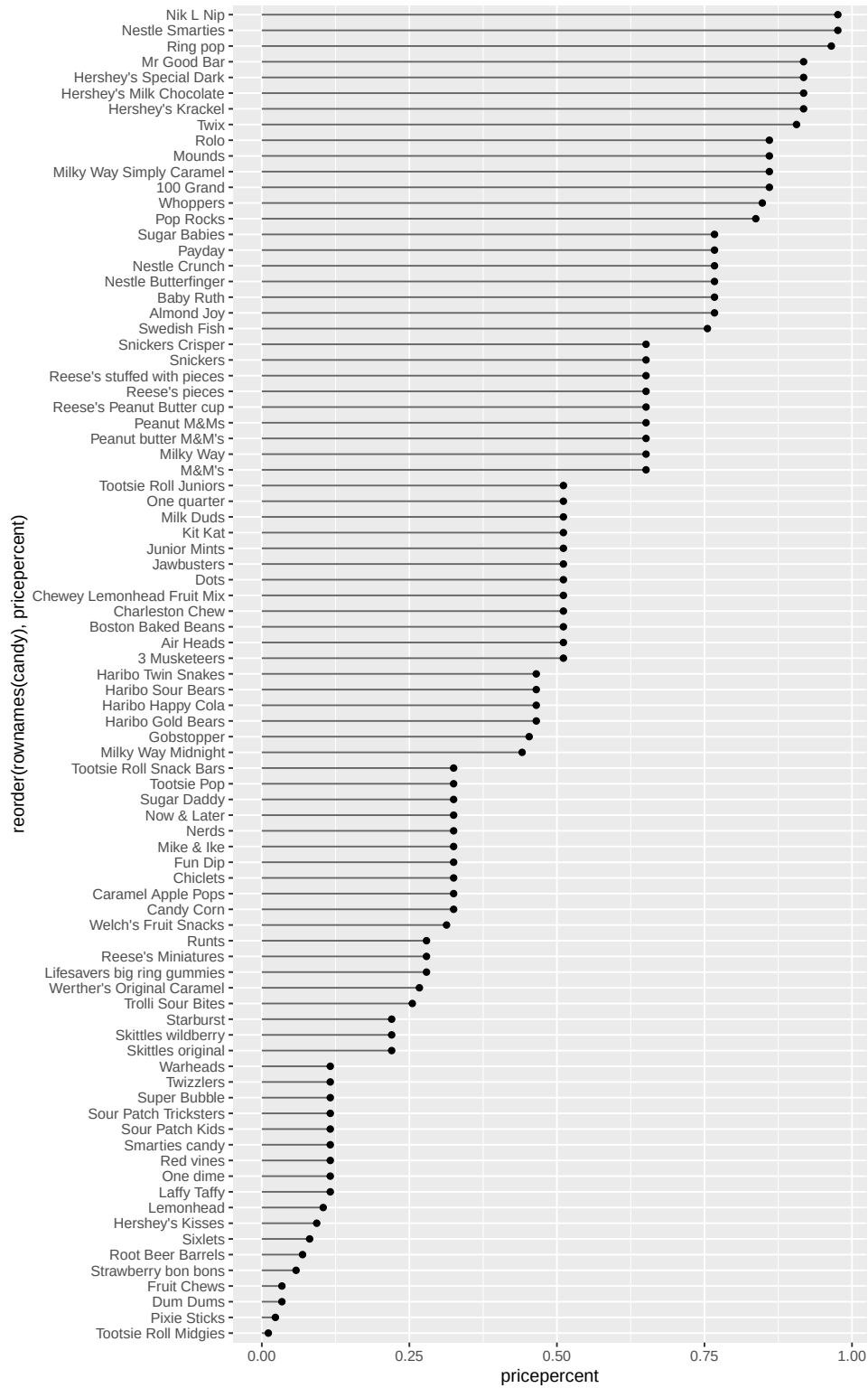
```

# Lollipop chart of pricepercent
ggplot(candy) +
  aes(x = pricepercent,
      y = reorder(rownames(candy), pricepercent)) +

  # stems of the lollipop
  geom_segment(aes(
    x = 0,
    xend = pricepercent,
    yend = reorder(rownames(candy), pricepercent)
  ), col = "gray40") +

  # dots of the lollipop
  geom_point()

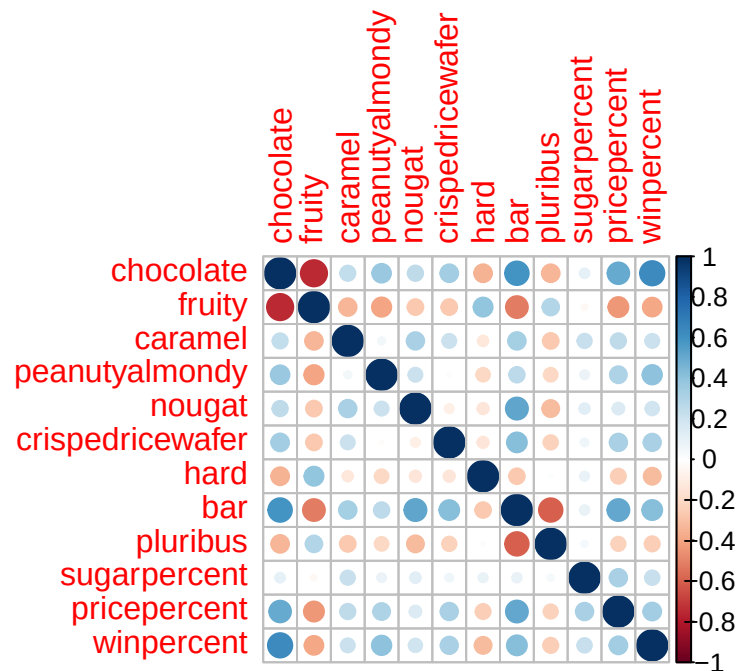
```



```
library(corrplot)
```

corrplot 0.95 loaded

```
candy_BK <- candy  
candy <- candy %>% dplyr::select(-value_score)  
  
cij <- cor(candy)  
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolate and fruity are the most strongly anti-correlated variables in the dataset.

Q23. Similarly, what two variables are most positively correlated?

Chocolate and winpercent are the most strongly positively correlated variables in the dataset.

```
# Run PCA with scaling  
pca <- prcomp(candy, scale = TRUE)
```

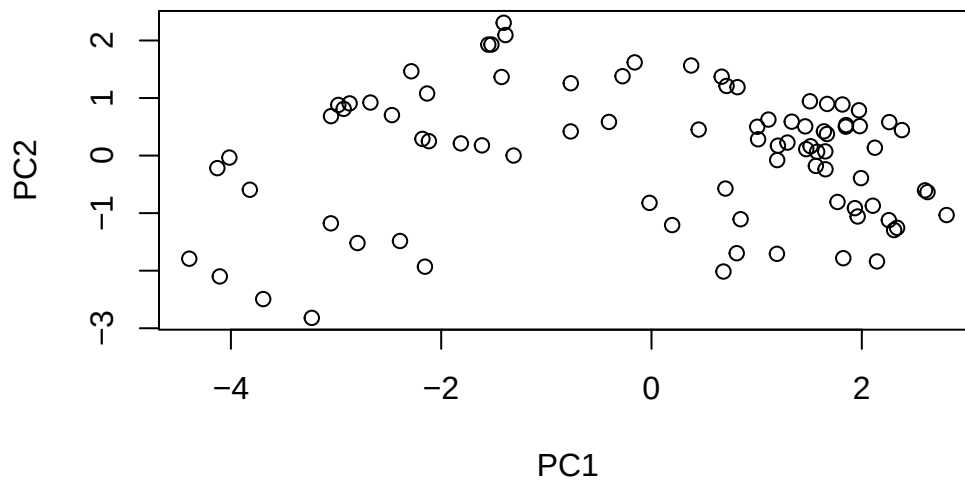
```
# Summary of PCA results
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
plot(pca$x[,1:2])
```



```
# Run PCA
pca <- prcomp(candy, scale = TRUE)

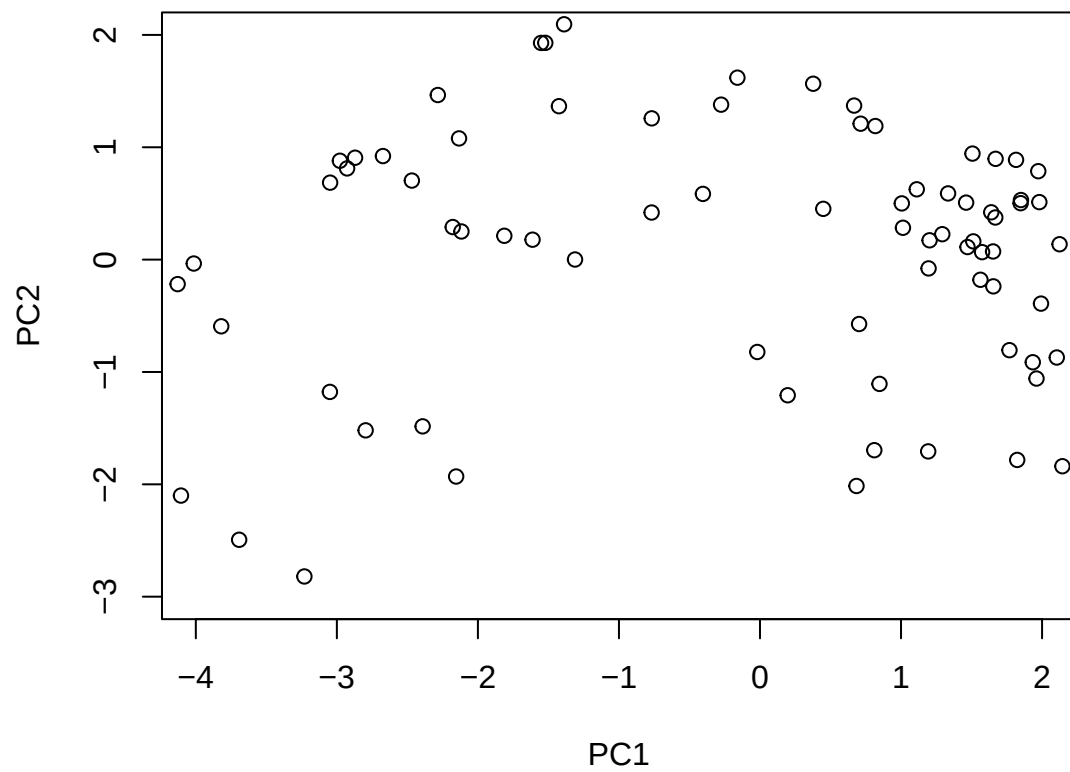
# Summary of PCA results
summary(pca)
```

Importance of components:

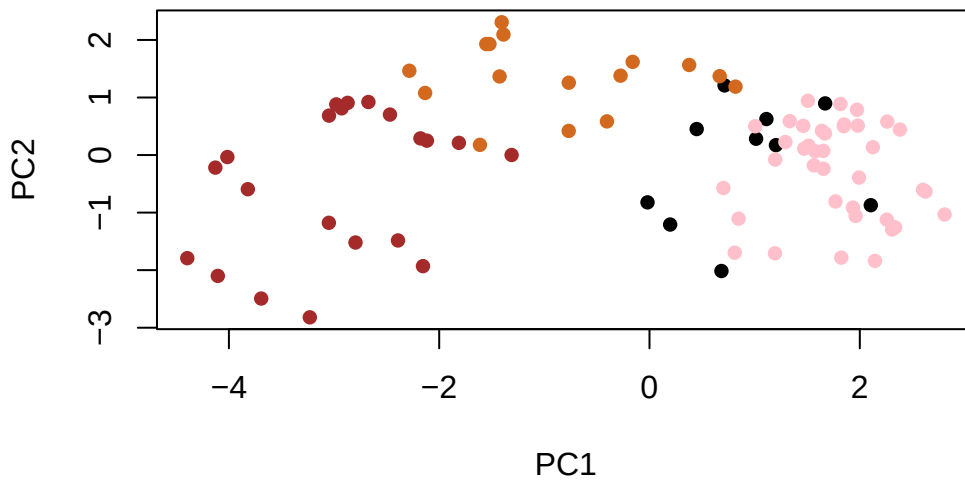
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
# Plot PC1 vs PC2 with custom axis limits
plot(pca$x[,1:2],
      xlim = c(-4, 2), # x-axis range
      ylim = c(-3, 2), # y-axis range
      xlab = "PC1",
      ylab = "PC2")
```



```
plot(pca$x[,1:2], col=my_cols, pch=16)
```

```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
```

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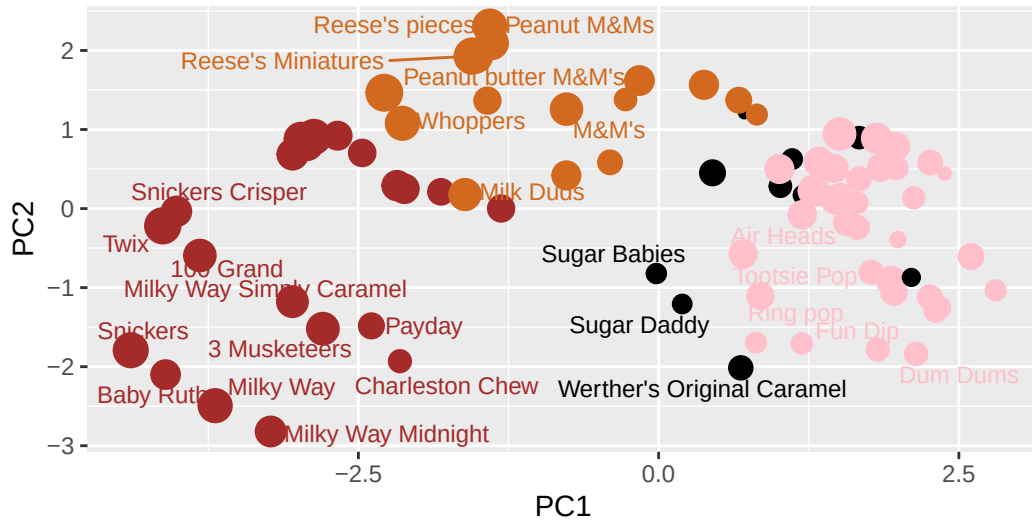
```
# PCA scatter plot (PC1 vs PC2)
p <- ggplot(my_data) +
  aes(x = PC1, y = PC2,
      size = winpercent / 100,
      text = rownames(my_data),
      label = rownames(my_data)) +
  geom_point(col = my_cols)
```

```
library(ggrepel)
```

```
p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
       caption="Data from 538")
```

Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

```
library(plotly)
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

last_plot

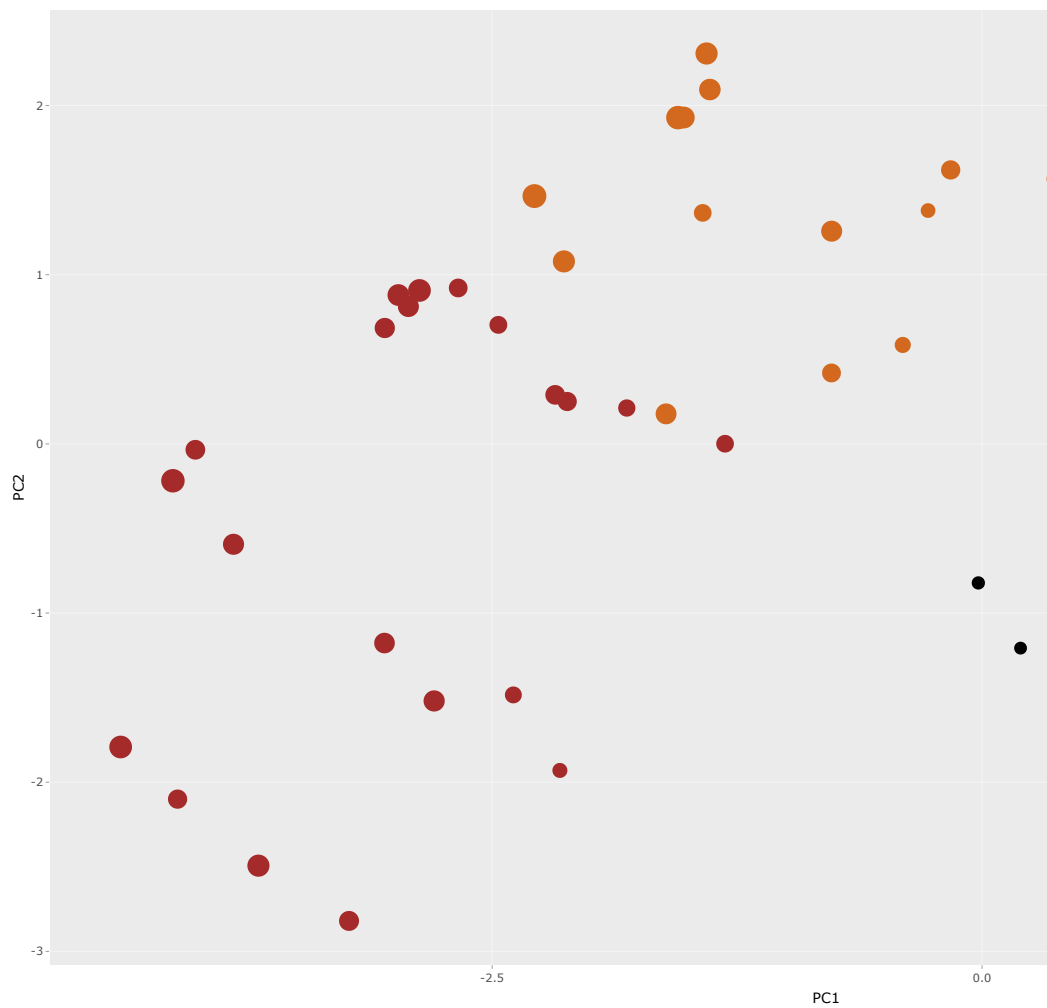
The following object is masked from 'package:stats':

filter

The following object is masked from 'package:graphics':

layout

```
ggplotly(p)
```



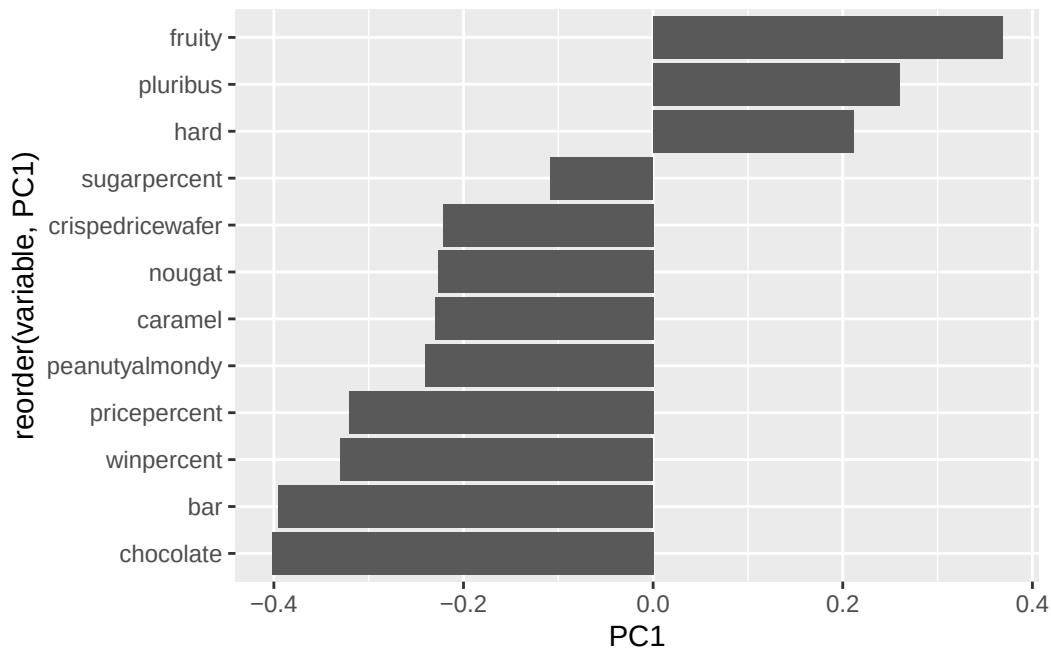
Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

PC1 is strongly driven in the positive direction by chocolate (and related bar-type variables), which matches what we previously saw in the correlation analysis where chocolate was positively associated with winpercent and negatively associated with fruity candy.

```
library(ggplot2)

# Put PCA rotation into a data frame with variable names
loadings <- as.data.frame(pca$rotation)
loadings$variable <- rownames(loadings)

# Bar plot of PC1 loadings
ggplot(loadings) +
  aes(x = reorder(variable, PC1), y = PC1) +
  geom_col() +
  coord_flip()
```



Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning”

candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A “winning” candy is typically chocolate-based, often in bar form, and not fruity, and this conclusion is consistently supported across visualization, correlation analysis, and PCA, which all show chocolate as the higher popularity.